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A hybrid A* path planning algorithm for underground load-haul-dump vehicles guided by reference states

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Abstract

As a critical enabler of trackless mining operations, load-haul-dump (LHD) vehicles rely heavily on efficient and safe path planning to support the industrial deployment of autonomous driving systems. However, existing planning algorithms often suffer from heavy computational demands, limited adaptability to the constrained kinematics of articulated LHDs, and insufficient robustness in complex underground environments. To address these challenges, we propose a Hybrid A* path planning method guided by high-level reference states. The method first integrates the A* algorithm with generalized Voronoi diagrams to rapidly generate a reference state set along tunnel centerlines. This reference serves as a structural guide for downstream search. Then, a region-aware dynamic node expansion strategy and a multi-objective heuristic cost function are introduced to improve planning efficiency and path quality. Comparative evaluations across three representative underground tunnel scenarios demonstrate that the proposed approach outperforms baseline methods in terms of safety, efficiency, and robustness, providing a viable solution for motion planning of LHDs in confined underground spaces.

Keywords: articulated vehicles, path planning, hybrid A*, generalized Voronoi diagram (GVD)

1. Introduction

As a cornerstone of trackless underground mining, load-haul-dump (LHD) vehicles have seen steady improvements in intelligence, largely fueled by advancements in autonomous driving technologies—and are now inching toward large-scale industrial adoption. At the heart of these systems lies the path planning module, which fundamentally shapes both navigation reliability and operational productivity of driverless LHDs.

Although a variety of path planning methods have been successfully applied in ground transportation, warehouse logistics, and robotic navigation, they typically lack tailored optimizations for the constrained, elongated, and unstructured roadway structures found in subterranean mines, as well as the distinctive kinematic constraints of LHDs. As a result, deploying these algorithms underground still poses significant difficulties. For instance, sampling-based planners like RRT* have shown promise in open environments but struggle with narrow tunnels and vehicle articulation.

Graph-based path planning algorithms such as Dijkstra [1] and A* [2] are among the most widely adopted approaches in underground mining scenarios, mainly due to their

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deterministic nature and ability to guarantee globally optimal solutions. However, these algorithms often suffer from excessive computational costs, low search efficiency, and inadequate real-time performance, especially when applied to large or complex maps. To address these limitations, several studies have explored the incorporation of swarm intelligence techniques—such as ant colony optimization (ACO) [3] and particle swarm optimization (PSO) [4]—to improve both path quality and planning efficiency, while still ensuring feasibility under constrained underground conditions.

In comparison, sampling-based path planning algorithms such as probabilistic roadmaps (PRMs) [5] and rapidly-exploring random trees (RRTs) [6] demonstrate higher search efficiency in high-dimensional spaces and complex, unstructured environments. However, these approaches typically generate paths with poor smoothness, face difficulties in efficient collision checking, and often lack stability across repeated trials. To address these shortcomings, various improvements have been proposed, including the adoption of dynamic step-size adjustment, optimal tree rewiring strategies [7], and Hybrid frameworks that combine the strengths of both PRM and RRT methods [8], aiming to enhance path quality and increase the overall success rate of planning.

In addition, swarm intelligence algorithms such as genetic algorithms [9], PSO [10], and ACO [11] have also been applied to path planning in underground mining due to their strong global search capabilities. However, these methods typically suffer from slow convergence and a tendency to fall into local optima, which limits their robustness in complex or dynamic environments. To overcome these limitations, researchers have proposed several improvements to the ACO algorithm, such as the use of pseudo-random proportional rules, retreat punishment mechanisms, and dynamic pheromone regulation strategies [12]. Other enhancements include entropy-increasing mechanisms [13] and initial pheromone distribution differentiation strategies [14], aiming to increase solution diversity and improve overall stability.

The rapid advancement of artificial intelligence has led to the increasing adoption of deep learning [15–17] and reinforcement learning [18–21] techniques in the field of path planning, including applications in unstructured environments such as underground mines. These learning-based approaches are capable of autonomously acquiring path optimization strategies from large-scale environmental perception data, showing promising adaptability and control over path diversity. However, their practical deployment remains limited by several factors, including the requirement for high-quality training datasets, limited generalization capabilities, and the heavy demand on computational resources.

Although the mentioned methods have laid a theoretical foundation and provided conceptual directions for path planning of underground LHD vehicles, they typically lack tailored adaptations to the vehicle's kinematic characteristics and the unique constraints of subterranean environments. Most of these approaches rely solely on discrete coordinate sets to generate paths, which limits their ability to model realistic maneuverability. In contrast, the Hybrid A* algorithm [22]

retains the heuristic-guided search framework while incorporating nonholonomic motion constraints, making the generated paths more consistent with the vehicle's actual kinematic behavior.

In recent years, the Hybrid A* algorithm has been widely applied in open-pit mines and other unstructured environments [23]. To accommodate complex terrain conditions or task-specific constraints, researchers have enhanced heuristic cost functions by incorporating multiple penalty terms—such as obstacle risk costs, collision-avoidance weighted distances [24], and traversability assessments [25]—to improve path safety and redundancy. By contrast, studies targeting underground mining environments or articulated vehicles remain relatively limited.

To address the global path planning challenges for articulated vehicles, various studies have explored tailored enhancements to the Hybrid A* algorithm. Ren *et al* [26] redesigned the forward expansion strategy and multi-objective cost function based on the operational characteristics of underground LHDs, leading to notable improvements in path quality within tunnel environments. Li *et al* [27] introduced Voronoi potential fields and a local map update mechanism, and further embedded multiple weighting factors into the cost function to achieve a balanced trade-off between path length and safety. Liu *et al* [28] employed bidirectional *Dubins* curves to avoid redundant turning points and impassable segments, while applying factor graph-based post-processing to shorten path redundancy and reduce the frequency of heading direction changes.

Furthermore, in pursuit of higher search efficiency, several studies have proposed novel node expansion strategies to accelerate Hybrid A* planning. These include backward search techniques [29], hierarchical map-based planning [30]—which uses coarse-resolution maps to guide expansion and fine-resolution maps to refine path quality—and bidirectional search mechanisms [31]. Collectively, these strategies aim to improve the real-time responsiveness of Hybrid A* in complex environments.

Although these enhancements have led to noticeable improvements in path quality and search performance, they generally fall short of fundamentally restructuring the search mechanism itself. The guiding effectiveness of the cost function remains limited, especially in large-scale or structurally complex tunnel environments, where the Hybrid A* still suffers considerable computational overhead and inefficient exploration. Additionally, the lack of dynamic clearance optimization in many of these studies can lead to wall-hugging trajectories, increasing the potential risk of scraping or collision in narrow tunnel passages.

To address the limitations of Hybrid A* in large-scale, complex tunnel environments—particularly its high computational cost and limited path safety—this study proposes a Hybrid A* path planning algorithm guided by reference states. The method is tailored for global path planning of articulated vehicles such as underground LHD machines. The main contributions are summarized as follows:

1. A fast construction method is developed for generating reference state sets by combining the generalized Voronoi diagram (GVD) and A*, which enables the division of the search space into two structured subregions: intersection zones and closed tunnels. This guides the subsequent planning process with spatial constraints.
2. A node expansion strategy is proposed based on reference state matching and dynamic switching, which performs focused search in closed areas and preserves native expansion in intersections, thereby reducing node redundancy and improving both planning efficiency and search stability.
3. A multi-objective heuristic cost function is designed that jointly considers path length, heading variation, articulation angle deviation, and safety margin, enabling a dynamic trade-off between feasibility and path quality.

The remainder of this paper is organized as follows. In section 2, we formalize the kinematic model of articulated vehicles and define the constrained-space path planning problem. Section 3 details the implementation of the proposed algorithm, including the construction of the reference state set, the optimization logic of node expansion strategies, and the design of the multi-objective heuristic cost function. Section 4 presents three typical underground tunnel scenarios to compare the proposed method with conventional approaches and their bidirectional search variants in terms of search efficiency and path quality. Finally, section 5 concludes the study, discussing the advantages and limitations of the proposed method and outlining directions for future work.

2. Problem formulation

2.1. Kinematic model of LHD

Compared with conventional wheeled steering vehicles, underground LHD machines adopt an articulated steering mechanism, where the turning motion is driven by the relative deflection between the front and rear bodies. This configuration forms a typical nonholonomic constraint system. In tunnel environments, path planning must ensure sufficient clearance between both vehicle sections and the tunnel walls, while also maintaining motion continuity and smoothness. As illustrated in figure 1, the unique kinematic characteristics of the LHD can be described as a relative rotational movement of the front and rear bodies around a central articulation joint. The definitions of the relevant physical variables are listed in table 1.

Assuming the LHD vehicle is moving forward and using the front axle center $O_f(x_f, y_f)$ as the reference point, the kinematics can be formulated via the following state-space equation [32]:

$$\begin{bmatrix} \dot{x}_f \\ \dot{y}_f \\ \dot{\theta}_f \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} \cos \theta_f & 0 \\ \sin \theta_f & 0 \\ \frac{\sin \gamma}{l_f \cos \gamma + l_r} & \frac{l_r}{l_f \cos \gamma + l_r} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \dot{\gamma} \end{bmatrix} \quad (1)$$

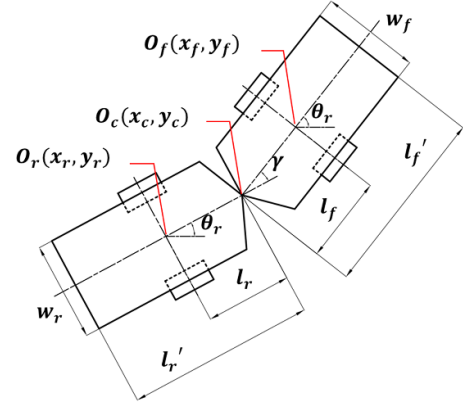


Figure 1. Kinematic model of an articulated vehicle.

Table 1. Notations and definitions of the kinematic model.

Symbol	Physical meaning
$O_c(x_c, y_c), O_f(x_f, y_f), O_r(x_r, y_r)$	Coordinates of articulation joint and the center of front and rear axles
l_f, l_r	Distance from front/rear axle to articulation joint
l_f', l_r'	Length of the front/rear body
w_f, w_r	Width of the front/rear body
θ_f, θ_r	Yaw angles of front/rear body
v_f, v_r	Linear velocity at front/rear axle center
γ	Articulation angle
$\dot{\gamma}$	Articulation angular velocity

where the state vector is $x = [x_f \ y_f \ \theta_f \ \gamma]^T$, representing the front-axle center position, heading angle, and articulation angle; the control input is $u = [v \ \dot{\gamma}]^T$ denoting forward velocity at O_f and articulation angular rate, respectively.

2.2. Path planning formulation

In this study, we perform global path planning for the underground LHD vehicle on a prior tunnel map constructed via SLAM. After preprocessing (e.g. binarization and filtering), a binary occupancy grid map BM is obtained at resolution δ . Each grid cell $BM_{i,j}$ is defined as follows (see equation (2)):

$$\begin{cases} BM_{i,j} = \begin{cases} 0, & \text{if cell } (i,j) \text{ is free} \\ 1, & \text{if cell } (i,j) \text{ is occupied} \end{cases} \\ X_{\text{Obs}} = \{(i,j) \mid BM_{i,j} = 1\} \\ X_{\text{Free}} = \{(i,j) \mid BM_{i,j} = 0\} \end{cases} \quad (2)$$

Here, the grid map has dimensions $m \times n$, with indices $i, j \in \mathbb{Z}^+$. X_{Obs} as the set of obstacle cells, X_{Free} as the set of free-space cells.

In a constrained underground environment, given a start pose, given a start pose $S(x_{\text{start}}, y_{\text{start}}, \theta_{\text{start}})$ and goal pose $G(x_{\text{goal}}, y_{\text{goal}}, \theta_{\text{goal}})$, the path planning must satisfy:

1. A continuous path $\sigma : [0, 1] \rightarrow X_{\text{Free}}$ that connects S and G , where θ_{start} and θ_{goal} denote the heading angles of the front body at the start and goal poses respectively;
2. The path must satisfy nonholonomic kinematic constraints of an articulated vehicle and accommodate bidirectional motion, while avoiding collisions between the vehicle bodies and tunnel walls;
3. The central articulation point $O_c(x_c, y_c)$ is used as the reference pose, ensuring continuity and differentiability of the vehicle's pose along the trajectory.

Thus, the path planning problem for the underground LHD can be formulated as the following optimization problem:

$$\min_{x(t), u(t)} J(\sigma) = \int_0^T C(\xi(t), u(t)) dt \quad (3)$$

$$s.t. \begin{cases} \xi(0) = S, \xi(T) = G \\ \|p(i)\| = v_{\text{ref}}, |\gamma(t)| \leq \gamma_{\text{max}}, \forall t \in [0, T] \\ \min_{i,j \in X_{\text{Obs}}} \|p(t) - (i,j)\|_2 \geq d_{\text{Safe}}, \forall t \in [0, T] \end{cases}$$

where:

The driving velocity is set to a constant value $\|p(i)\| = v_{\text{ref}}$; $|\gamma(i)| \leq \gamma_{\text{max}}$, constrains the articulation angular velocity to be below its maximum $\dot{\gamma}_{\text{max}}$;

$|\gamma(t)| \leq \gamma_{\text{max}}$, ensures that the articulation angle does not exceed its physical limit γ_{max}

The distance between $p(t)$ and any obstacle is always greater than the minimum safety margin d_{Safe} ;

3. Path planning method

This study proposes a Hybrid A* path planning algorithm guided by high-level reference states, specifically designed for global navigation of underground LHD vehicles in narrow tunnel environments. The core innovation lies in the integration of the A* algorithm with the GVD to construct a reference state set aligned with tunnel centerlines, which provides structured guidance for node expansion. Furthermore, by optimizing the node expansion strategy and refining the heuristic cost function, the algorithm improves search efficiency by suppressing redundant node generation—while maintaining stability and path safety. The following sections detail the algorithmic implementation and present experimental validation under representative underground mining scenarios.

3.1. Upper-level reference state guidance set

3.1.1. Skeleton extraction. The GVD extends the traditional Voronoi diagram by partitioning the free space of the map into a set of boundaries, each formed by points equidistant to neighboring obstacles. This skeletal framework, similar to other distance field construction methods [33, 34], provides a path that maximizes the clearance between the robot and obstacles, allowing navigation along routes that are as far from

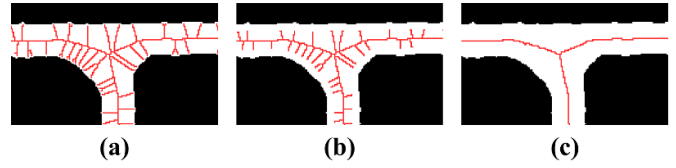


Figure 2. Practical pruning results.

obstacles as possible. In this study, the GVD is rapidly generated on the binary occupancy grid map constructed via SLAM using the Brush Fire algorithm. By simulating the propagation of ‘flames’ from obstacle boundaries into free space, this algorithm efficiently constructs equidistant boundaries and is both simple to implement and computationally effective [35]. However, as shown in figure 2(a), the resulting GVD still contains a substantial number of redundant branches.

These redundant branches not only slow down the search process but also form sharp angles at intersections, which degrades path smoothness. Moreover, some branches are too close to obstacle boundaries, increasing the risk of collision and possibly leading to infeasible paths. To address these issues, this study proposes a fast branch-pruning algorithm for the GVD tailored to tunnel environments. Let the obstacle set be X_{Obs} , and the generated GVD be $V(X_{\text{Obs}})$, which is defined as follows:

- (1) Voronoi edge $ve(c_i, c_j)$: the set of Voronoi edges with endpoint $c_i, c_j \in X_{\text{Obs}}$;
- (2) Voronoi vertex $vv(v_i, v_j)$: the Voronoi vertex connecting two Voronoi edges ve_i and ve_j ;
- (3) Node degree $ND(vv_i)$: the number of Voronoi edges incident to vv_i ;
- (4) Safety margin $D_{\text{Safe}}(p_i) = \min_{c \in X_{\text{Obs}}} \|c - vv_i\|_2$: the Euclidean distance between any location $p_i(x_i, y_i)$ and the nearest obstacle.

Moreover, the classification of Voronoi vertices^{34]} is as follows (equation (4)):

$$ND(vv_i) = \begin{cases} 1 \Rightarrow & \text{end point, the terminal of a branch,} \\ 2 \Rightarrow & \text{link point, located within a branch,} \\ \geq 3 \Rightarrow & \text{meet point, the intersection of branches.} \end{cases} \quad (4)$$

Based on the above definitions, a Voronoi branch can be described as a sequence of consecutive Voronoi nodes starting from an end point and terminating at the nearest intersection point. The proposed fast branch-pruning algorithm proceeds as illustrated in figure 3.

- (1) As shown in figure 3(b), set a safety margin threshold D_T (at least half of the vehicle's maximum width, because the planning reference point lies along the vehicle's longitudinal centerline) and remove Voronoi vertices vv_i with safety margin less than the threshold ($D_{\text{Safe}}(vv_i) < D_T$),

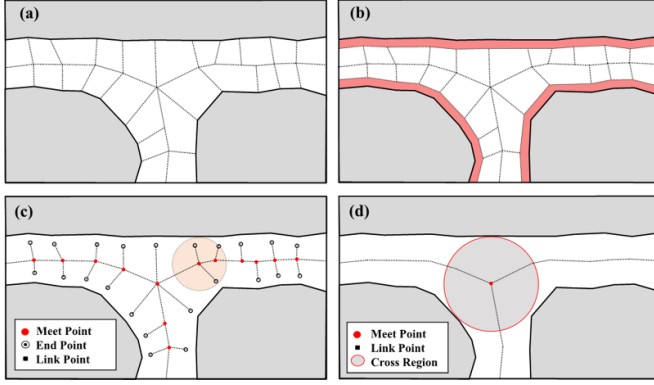


Figure 3. Efficient pruning algorithm.

to eliminate potential collision risks. The actual effect is shown in figure 2(b);

- (2) As shown in figure 3(c), traverse all end points $\{vv_i | ND(vv_i) = 1\}$ and search along adjacent edges until an intersection or another end point is reached, thereby forming the candidate branch set $B = \{B_1, B_2, \dots, B_N\}$, where:

$$B_i = \{vv_1, vv_2, \dots, vv_k | ND(vv_1) = 1, ND(vv_i) = 2 (i < k), ND(vv_k) > 2\} \quad (5)$$

- (3) For each branch B_i , use its associated intersection vv_m as the center and the safety margin $D_{Safe}(vv_m)$ as the radius to construct a circular region:

$$A_m = \{p \in \mathbb{R}^2 | \|p - vv_m\|_2 \leq D_{Safe}(vv_m)\}. \quad (6)$$

If $\forall vv_i \in B_i, vv_i \subset A_m$, then branch B_i is considered redundant and removed;

- (4) Repeat steps (2) and (3) until no new redundant branches are generated. The resulting skeleton nodes after pruning are illustrated in figure 2(c).

After the branch pruning process, the resulting map skeleton, as shown in figure 3(d), contains meet points vv_M only at the intersections of multiple branches. For each meet point vv_M , a circular region A_m is defined with the junction as the center and the safety margin $D_{Safe}(vv_M)$ as the radius. These intersection regions serve as references for adjusting node expansion strategies in subsequent planning steps.

3.1.2. Generation of reference state set. To reduce the computational complexity of the Hybrid A* algorithm in tunnel-like environments, this study leverages the pruned skeleton topology for guiding the generation of the reference state set via the following steps:

- (1) On the skeleton graph, select the Voronoi nodes vv_s and vv_g closest to the start S and goal G , respectively. Use the

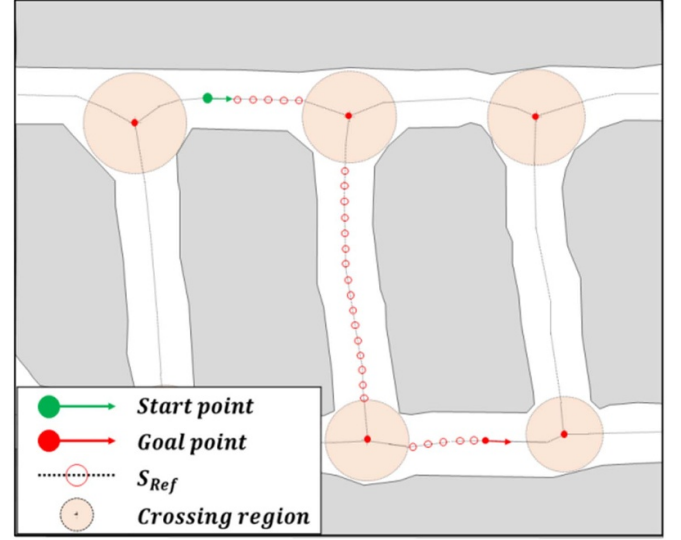


Figure 4. Generation of reference state set.

A* algorithm on the skeleton to generate an initial guiding path:

$$P_{guide} = \{vv_s, vv_i, \dots, vv_j, vv_g\} \quad (7)$$

- (2) Perform discrete sampling along the guiding path P_{guide} and remove nodes that fall within intersection regions A_m . The guiding path is thus partitioned into multiple curve segments $S_g \{s_{g1}, s_{g2}, \dots, s_{gp}\}$;
- (3) Each segment in the set S_g is smoothed and discretized at a fixed interval Δs , resulting in the reference state set S_{Ref} (as shown in figure 4), which provides dynamic guidance constraints for the Hybrid A* algorithm under various scenarios. For each segment, the order is defined as forward if the initial node points to the terminal node of the current segment, otherwise as reverse:

$$\forall s_i \{vv_1, \dots, vv_{p+q}\} \in S_g, \min_{i,j < (p+q)} \|vv_i - vv_j\|_2 > \Delta s \quad (8)$$

3.2. Lower-level path planning algorithm

To address the specific requirements of path planning for underground LHD vehicles, this study proposes an improved Hybrid A* algorithm that incorporates the upper-level guidance constraints. By enhancing the node dynamic expansion strategy and optimizing the heuristic cost function within the traditional Hybrid A* framework, the proposed approach significantly improves planning performance in narrow tunnel scenarios. The detailed procedure is illustrated in algorithm 1.

3.2.1. Forward expansion of nodes. As illustrated in figure 5, unlike the classic A* method that constructs the search space based on discrete grid cells, our approach uses the central articulation point of the LHD vehicle as the reference

Algorithm 1. Advanced Hybrid A* algorithm.**Input:** Start node S , Goal node G , Binary map $BM_{i,j}$, Reference State Segs S_{Ref} , Cross Region Set A_m **Output:** Feasible Path P

```

1.   $OpenSet \leftarrow \emptyset$  (priority queue for unevaluated nodes)
2.   $CloseSet \leftarrow \emptyset$  (set for evaluated nodes)
3.  Set  $f(S) \leftarrow h(S)$ , where  $h(S)$  is the heuristic cost from  $S$  to  $G$ 
4.  Add  $S$  to  $OpenSet$ 
5.  while  $OpenSet \neq \emptyset$ , do
6.      Remove node  $X_{current}$  with minimum cost from  $OpenSet$ 
7.      Add  $X_{current}$  to  $CloseSet$ 
8.      if  $distance(X_{current}, G) \leq d_{Analytic}$ , then
9.          Compute bidirectional Dubins curve  $c_{Dubins}$  from  $X_{current}$  to  $G$ 
10.         if  $c_{Dubins}$  is collision-free, then
11.             Backtrack parent nodes to construct  $P$ 
12.             return  $P$ 
13.         end if
14.     end if
15.     Set successor nodes  $N_{succ} \leftarrow \emptyset$ 
16.     Generate successor nodes  $N_{succ}$  with advanced expand strategy and insert  $N_{shot}$  into  $N_{succ}$ 
17.     if  $N_{succ}$  is empty, then
18.         Generate successor nodes  $N_k$  via kinematically feasible motions
19.         Insert  $N_k$  into  $N_{succ}$ 
20.     end if
21.     for each  $X_n \in N_{succ}$ , do
22.         if  $X_n \in CloseSet$ , continue
23.         Update accumulated cost with advanced function  $g(X_n) \leftarrow g_a(X_{current}) + cost(X_{current}, X_n)$ 
24.         Update  $h(X_n)$  with advanced heuristic cost function  $h_a(X_n, G)$ 
25.         Calculate total cost  $f(X_n) \leftarrow g(X_n) + h(X_n)$ 
26.         if  $X_n \notin OpenSet$ , then
27.             Set  $X_n.parent \leftarrow X_{current}$ 
28.             Insert  $X_{current}$  into  $OpenSet$ 
29.         else if  $g(X_n) < existing\ g_{old}(X_n)$ , then
30.             Update  $g_{old}(X_n) \leftarrow g(X_n)$ 
31.              $X_n.parent \leftarrow X_{current}$ 
32.         end if
33.     end for
34. end while
35. return failure (no path found)

```

and employs a fixed linear velocity v_{ref} and a set of discretized articulation angular rates:

$$\dot{\gamma}_{Ref} \{ \dot{\gamma}_i = -\dot{\gamma}_{max} + i\Delta\dot{\gamma} | i \in \mathbb{Z}^+, |\dot{\gamma}_i| \leq \dot{\gamma}_{max} \} \quad (9)$$

as control inputs. Nodes are annotated with driving direction to support bidirectional motion. By integrating the vehicle's kinematic equations, node expansion is driven in a way that preserves full vehicle pose, resulting in curvature-continuous and nonholonomic-compliant smooth paths. Additionally, an oriented bounding box based collision detection mechanism is introduced to prune any expansion that may lead to collisions, ensuring path safety and physical feasibility:

3.2.2. Dynamic node expansion strategy. The node expansion process in the Hybrid A* algorithm can be divided

into two stages [36]: the first stage is forward expansion, in which continuous paths are generated based on vehicle kinematic constraints; the second stage is analytic expansion, where a *Dubins* curve or Reeds – Shepp curve is employed to quickly connect the current node to the goal state, ensuring that the path precisely terminates at the desired pose. Since underground LHD vehicles cannot fully execute certain motion primitives of the Reeds – Shepp solution, this study utilizes *Dubins* curves for the analytic expansion.

In the forward expansion stage, we propose a dynamic node expansion strategy: before generating discrete nodes using the kinematic equations, a *Dubins* curve is computed to guide the current node toward the reference state provided by the upper-level planner. This approach significantly improves both search efficiency and path quality. The detailed procedure is as follows:

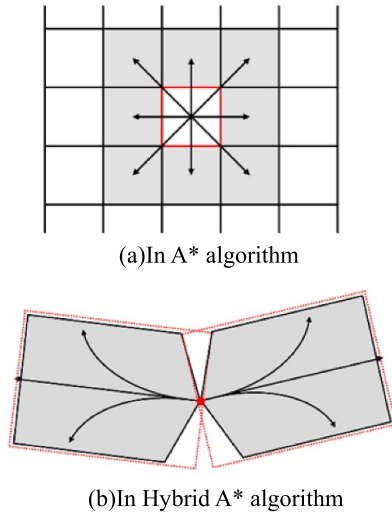


Figure 5. Comparison of node expansion strategies.

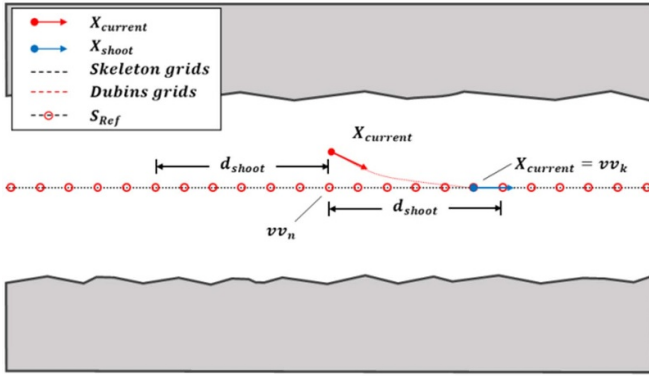


Figure 6. Shooting strategy for tunnel segments.

1. Define the traversal radius d_{shoot} for rapid *Dubins* curve connection;
2. If the current node X_{current} is in a single tunnel but does not belong to any reference segment i.e., $s_i \in S_{\text{Ref}}, X_{\text{current}} \cup s_i = \emptyset$, and for all $vv_m \in A_m, \|X_{\text{current}} - vv_m\|_2 > d_{\text{shoot}}$ then as shown in figure 6, select the reference segment $s_n \in S_{\text{Ref}}$ closest to X_{current} . Starting from the nearest point vv_n , iteratively select nodes vv_n within d_{shoot} of X_{current} along both ends of the segment, and compute the curve $c_{\text{shot}}(X_{\text{current}}, vv_n)$. If c_{shot} passes collision checking, stop traversal and add $X_{\text{shoot}} = vv_k$ to the successor node set N_{succ} . If X_{current} connects to X_{shoot} in the forward direction ($k > n$) of the reference segment, set its expansion as forward; otherwise, set as reverse.
3. If the current node is located in an intersection region $a_m \in A_m$, but does not belong to any reference segment (as shown in figure 7), traverse all reference segments s_i incident to a_m . Starting from the endpoint vv_T closest to the intersection, select nodes vv_K within a distance d_{shoot} from vv_T and solve for the *Dubins* curve $c_{\text{shot}i}$ between the current node and vv_K . If $c_{\text{shot}i}$ passes collision checking, add node $X_{\text{shoot}} = vv_k$ to the successor node set N_{succ} . Meanwhile, depending on the traversal direction along the

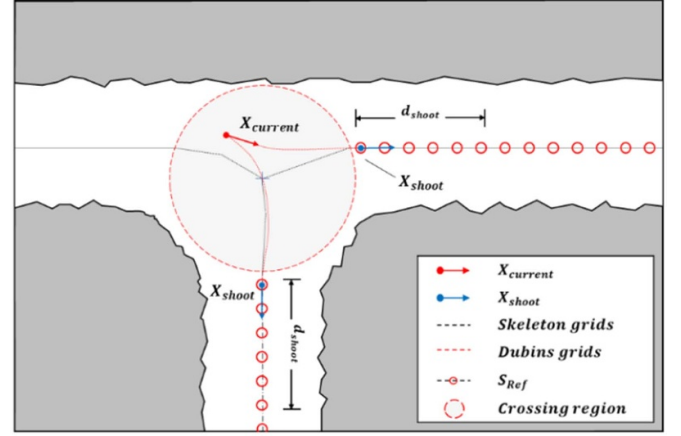


Figure 7. Shooting strategy for intersection regions.

segment (from the initial point to vv_K), set the expansion direction as forward; otherwise, if the traversal starts from the segment's final point, set the direction as reverse.

4. If the current node $X_{\text{current}} = vv_i$ lies on a reference segment s_i , then, according to the expansion direction, add the adjacent next node vv_j to the successor node set, N_{succ} where:

$$vv_j = \begin{cases} vv_{i+1}, & \text{forward expansion,} \\ vv_{i-1}, & \text{reverse expansion.} \end{cases} \quad (10)$$

5. If the current node neither belongs to any reference segment nor can be connected to any segment by a collision-free curve, then the successor nodes are generated directly via the kinematic model.

3.2.3. Multi-objective heuristic cost function. In the Hybrid A* algorithm, the cost function is built upon the framework $f(x) = g(x) + h(x)$ integrating vehicle kinematic constraints. However, conventional approaches have not fully adapted the cost formulation for articulated vehicles. In this study, the cost function is further optimized by incorporating both the mechanical characteristics of articulated vehicles and the constraints of narrow tunnel environments, thus achieving scene-specific optimization for underground mine path planning. The improved cost function is formulated as follows:

$$g(X_{\text{succ}}) = g(X_{\text{current}}) + l_s + \lambda_1 \cdot |\Delta\gamma| + \lambda_2 \cdot |\gamma| + w_{\text{rev}} + \frac{\lambda_3}{D_{\text{Safe}}(X_{\text{succ}})} \quad (11)$$

where:

1. Displacement penalty:

$$l_s = \begin{cases} \alpha \|X_{\text{succ}} - X_{\text{current}}\|_2, & \alpha \in (0, 1), X_{\text{succ}} \in S_{\text{Ref}}, \\ \|X_{\text{succ}} - X_{\text{current}}\|_2, & X_{\text{succ}} \notin S_{\text{Ref}}. \end{cases} \quad (12)$$

Here, the reference state incentive factor α ensures that, when optimizing for global path length, the current node is encouraged to connect to the upper-level reference state.

2. Steering dynamics penalty: The penalty coefficient λ_1 is multiplied by the absolute articulation increment $\Delta\gamma = \gamma_{\text{succ}} - \gamma_{\text{current}}$ to suppress system oscillations in the articulation steering and improve path smoothness;

3. Articulation stability penalty: The penalty coefficient is multiplied by the absolute value of the articulation angle $|\gamma|$, discouraging extreme steering angles that could reduce path smoothness;

4. Reverse motion penalty:

$$w_{\text{rev}} = \begin{cases} 0, & \vec{v}_{\text{current}} \cdot \vec{v}_{\text{succ}} > 0, \\ \lambda_{\text{rev}}, & \vec{v}_{\text{current}} \cdot \vec{v}_{\text{succ}} = 0. \end{cases} \quad (13)$$

This term encourages the algorithm to maintain unidirectional travel, reducing the risk of unsafe U-turns for the LHD in narrow tunnels;

5. Safety margin penalty: The penalty coefficient λ_3 is inversely proportional to the safety margin $D_{\text{Safe}}(X_{\text{succ}})$ of the successor node, encouraging the path to stay away from obstacles and thus improving feasibility.

To avoid local convergence caused by a single objective, the heuristic cost $h(X_{\text{succ}})$ for successor node X_{succ} adopts a multi-criteria selection strategy, providing direction for forward node expansion:

$$h(X_{\text{succ}}) = \max(\|X_{\text{succ}} - G\|_2, l_{\text{dubins}}(X_{\text{succ}}, G)) \quad (14)$$

(1) **Euclidean distance cost** $\|X_{\text{succ}} - G\|_2$: Computes the Euclidean distance between the successor node and the goal, providing an initial expansion reference.

(2) **Unconstrained Dubins cost** $l_{\text{dubins}}(X_{\text{succ}}, G)$: Ignores environmental obstacles, and uses the *Dubins* curve to estimate the shortest feasible path length from the current node to the goal.

Thus, the total expansion cost for successor node n_i is given by:

$$f(X_{\text{succ}}) = g(X_{\text{succ}}) + h(X_{\text{succ}}). \quad (15)$$

4. Experiment

4.1. Experimental design

To systematically evaluate the hierarchical path planning algorithm proposed in this study, a specific model of underground LHD vehicle is used as the research object. Typical underground mining scenarios are considered. By comparing the proposed approach with the conventional Hybrid A* algorithm adapted for LHD kinematic constraints and the bidirectional Hybrid A* algorithm, the overall advantages of the

Table 2. Geometric parameters of the underground LHD.

Parameter	Value	Parameter	Value
Front body length L_f (m)	5.0	Rear body length L_r (m)	5.0
Front body width W_f (m)	2.7	Rear body width W_r (m)	2.7
Distance from front axle to joint l_f (m)	2.0	Distance from rear axle to joint l_r (m)	2.0
Maximum articulation angle γ_{max} (rad)	$\frac{\pi}{6}$	Max articulation rate $\dot{\gamma}_{\text{max}}$ (rad s ⁻¹)	$\frac{\pi}{12}$
Reference velocity v_{ref} (m s ⁻¹)	2.0		

new method are comprehensively analyzed. The physical parameters of the LHD vehicle used in the experiments are summarized in table 2.

To ensure statistical consistency and comparability of the path quality evaluation metrics, all three path planning algorithms compared in the experiments adopt the same multi-objective heuristic cost function. Moreover, the raw paths generated by each algorithm are uniformly smoothed using cubic spline interpolation and resampled at equal intervals. This preprocessing eliminates the potential influence of non-uniform node distributions on the evaluation metrics, providing a unified data foundation for the subsequent path quality analysis.

All experiments were conducted on an Ubuntu 20.04 system (Intel Core i7 7820X processor, 48GB RAM) using the ROS framework. The experimental scenarios are selected from different areas of a typical underground mine map (figure 8), which are further divided into three representative types of tunnel structures. For each scenario, a fixed start point and various goal points are chosen for path planning experiments. Each start-goal pair is tested 20 times, and the pair with the maximum spatial separation is selected as the representative route for in-depth analysis of path-related parameters. As illustrated, the binary map (BM) used in the experiments has a resolution of $\delta = 0.2\text{m}$ and a grid size of 3080×970 ($x \in [0, 616]$, $y \in [0, 194]$).

1. Single tunnel scenario: A branch-free, closed tunnel segment within the region $x \in [0, 200]$, $y \in [105, 135]$ of the BM is selected as the experimental scenario (see figure 9). The start point is fixed at $S(40.50116.50, 0.0)$, and multiple goal points are placed along the tunnel centerline at 10 m intervals.

Multi-branch tunnel scenario: A complex tunnel section containing multiple branch paths is selected within the region $x \in [280, 320]$, $y \in [20, 180]$ of the BM as the experimental scenario (see figure 10). The start point is fixed at $S(298.05, 152.46, -1.57)$, and goal points are set along the centerline of each branch tunnel.

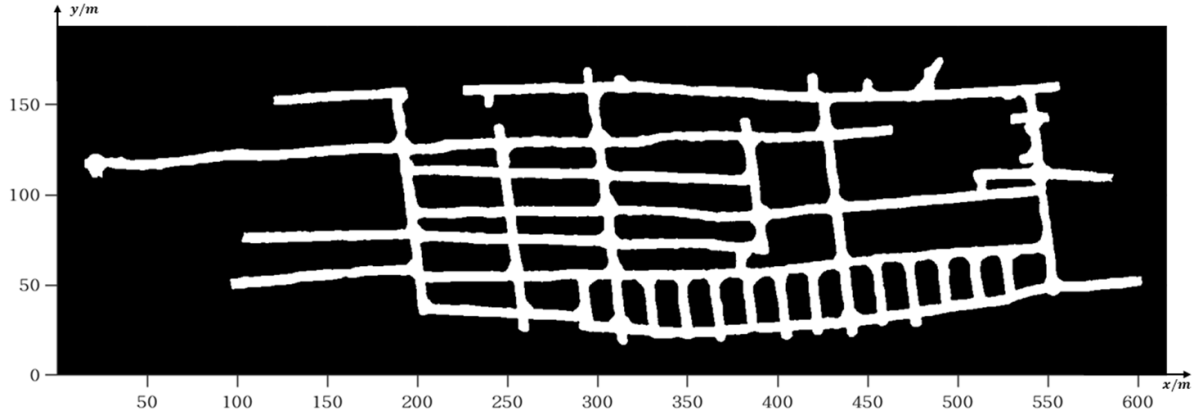


Figure 8. Global map of typical experimental scenarios.

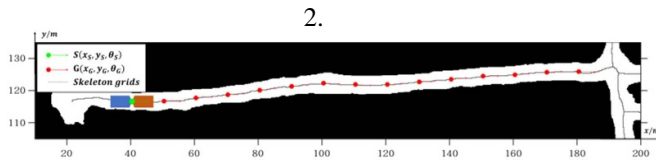


Figure 9. Single tunnel scenario.

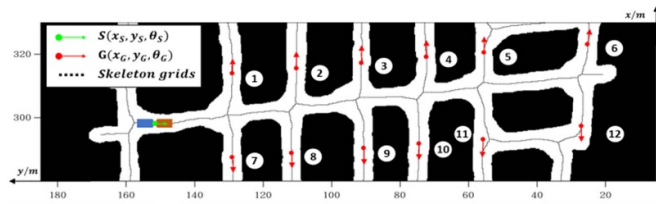


Figure 10. Multi-branch tunnel scenario.

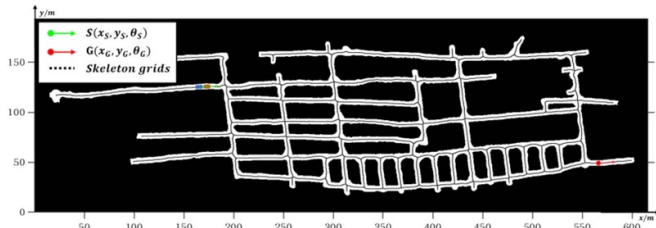


Figure 11. Grid-like tunnel scenario.

3. Grid-like tunnel scenario: A pair of start and goal points with large spatial separation is selected in the BM to represent a grid-like tunnel scenario (see figure 11). The start point is set at $(170.52, 125.74, 0.02)$, and the goal point at $G(566.28, 49.11, 0.02)$.

The parameters of the cost function for the above experiments are shown in table 3.

4.2. Experimental results and analysis

This section provides a comprehensive evaluation of the three methods in typical tunnel scenarios, focusing on metrics such as path length, safety margin, search efficiency and road

Table 3. Parameter coefficients of the heuristic cost function.

Parameter	Value	
	Local	Global
Reference state incentive α	0.3	0.3
Articulation stability penalty λ_2	0.2	0.2
Reverse motion penalty λ_3	2.0	1.0
Rapid connection range d_{shoot} (m)	4.0	4.0
Steering dynamics penalty: λ_1	0.5	0.25
Reverse motion penalty w_{rev}	0.5	0.25
Euclidean distance cost d_{Analytic} (m)	6.5	6.5

smoothness. In the tables, the baseline Hybrid A* algorithm is denoted as HA, the bidirectional Hybrid A* algorithm as BHA, and the proposed hierarchical Hybrid A* algorithm as HHA.

Path length, defined as the total travel distance of the generated path, is commonly used to quantify the performance of path planning algorithms. In practical applications, a shorter path length contributes to reduced travel time and lower energy consumption, making it a key performance indicator for evaluating planning efficiency [37]. As shown in figures 12(a) and (b), the three methods produce feasible paths of similar lengths (within 2 m difference) in the single-tunnel and multi-branch tunnel scenarios, due to the relatively fixed corridor geometry. However, in the grid-like tunnel scenario, the routes selected by different algorithms diverge significantly. As illustrated in figure 12(c), the proposed method reduces the total path length by over 12 m compared to both the baseline Hybrid A* and the bidirectional Hybrid A*, demonstrating superior path optimality.

Safety margin refers to the minimum distance between any point along the planned path and surrounding obstacles, and it is widely used to evaluate the safety and robustness of

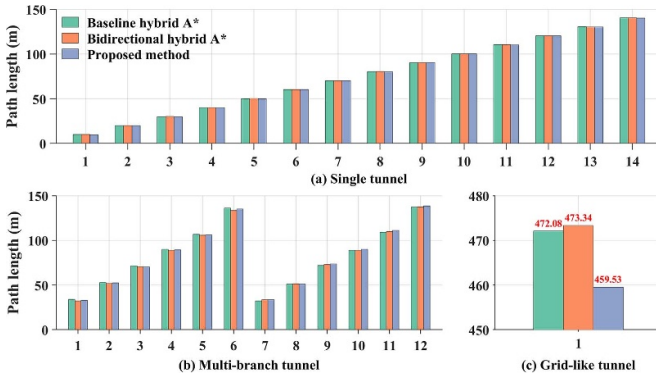


Figure 12. Path lengths.

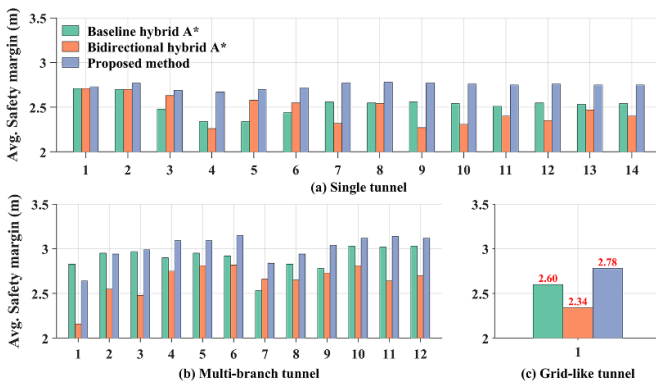


Figure 13. Average safety margins.

path planning algorithms [38]. In underground LHD operations, where tunnel width is highly constrained and the clearance between the vehicle body and tunnel walls is often less than 1 m, even minor deviations may lead to collision risks. Therefore, analyzing the safety margin is critical for assessing the safety assurance capabilities of path planning algorithms in such confined environments.

A comparative analysis of the average safety margins across different tunnel scenarios (as shown in table 4–6 and figure 13) reveals that the proposed method consistently achieves higher average safety margins than both the baseline Hybrid A* and the bidirectional Hybrid A* algorithms in most goal configurations. Moreover, the proposed method exhibits lower variance and higher stability in safety performance. Compared to the bidirectional Hybrid A*, which shows the weakest performance, the proposed approach provides over 0.3 m of additional safety clearance on average across all scenarios, and the improvements are more pronounced in more complex environments.

Further examination of the safety margin distributions along representative paths in different scenarios (see figures 14–16) indicates that the proposed algorithm produces safety margins that are skewed toward higher values, with greater median and mean values and tighter distribution. In contrast, the baseline and bidirectional Hybrid A* algorithms exhibit lower and more dispersed safety margins. These results suggest that the proposed approach offers superior safety

Table 4. Average safety margin in the single-tunnel scenario.

Index	Average safety margin (m)		
	HA	BHA	HHA
1	2.71 ± 0.09	2.71 ± 0.09	2.73 ± 0.09
2	2.70 ± 0.17	2.70 ± 0.10	2.77 ± 0.12
3	2.48 ± 0.23	2.63 ± 0.16	2.69 ± 0.18
4	2.34 ± 0.32	2.26 ± 0.38	2.67 ± 0.17
5	2.34 ± 0.33	2.58 ± 0.20	2.70 ± 0.16
6	2.44 ± 0.26	2.55 ± 0.31	2.72 ± 0.17
7	2.56 ± 0.32	2.32 ± 0.40	2.77 ± 0.21
8	2.55 ± 0.25	2.54 ± 0.33	2.78 ± 0.20
9	2.56 ± 0.25	2.27 ± 0.42	2.77 ± 0.20
10	2.54 ± 0.26	2.31 ± 0.42	2.76 ± 0.20
11	2.51 ± 0.26	2.40 ± 0.29	2.75 ± 0.19
12	2.55 ± 0.24	2.35 ± 0.32	2.76 ± 0.18
13	2.53 ± 0.24	2.47 ± 0.23	2.75 ± 0.18
14	2.54 ± 0.24	2.40 ± 0.32	2.75 ± 0.17

Note: Bolded mean values indicate a higher safety margin, while bolded standard deviations denote a more stable distribution of safety margins.

Table 5. Average safety margin in the multi-branch tunnel scenario.

Index	Average safety margin (m)		
	HA	BHA	HHA
1	2.83 ± 0.56	2.16 ± 0.45	2.64 ± 0.31
2	2.96 ± 0.79	2.55 ± 0.97	2.94 ± 0.77
3	2.97 ± 0.94	2.48 ± 0.91	2.99 ± 0.75
4	2.90 ± 0.82	2.75 ± 0.92	3.09 ± 0.82
5	2.94 ± 0.87	2.81 ± 0.88	3.09 ± 0.78
6	2.92 ± 0.84	2.82 ± 0.81	3.15 ± 0.70
7	2.53 ± 0.29	2.66 ± 0.53	2.84 ± 0.38
8	2.83 ± 0.85	2.65 ± 0.70	2.94 ± 0.79
9	2.77 ± 0.82	2.73 ± 0.81	3.04 ± 0.70
10	3.03 ± 0.85	2.81 ± 0.91	3.12 ± 0.77
11	3.02 ± 0.79	2.64 ± 0.78	3.14 ± 0.73
12	3.03 ± 0.77	2.70 ± 0.82	3.12 ± 0.70

Note: Bolded mean values indicate a higher safety margin, while bolded standard deviations denote a more stable distribution of safety margins.

Table 6. Average safe margin in the grid-like tunnel scenario.

Index	Average safety margin (m)		
	HA	BHA	HHA
1	2.60 ± 0.64	2.34 ± 0.55	2.78 ± 0.31

Note: Bolded mean values indicate a higher safety margin, while bolded standard deviations denote a more stable distribution of safety margins.

buffer control for underground LHDs, thereby effectively reducing collision risks in constrained tunnel environments.

During the path planning process, **searching duration** serves as a direct measure of an algorithm's computational overhead and real-time responsiveness in practical scenarios [39]. In this study, each of the three algorithms—baseline Hybrid A*, bidirectional Hybrid A*, and the proposed method—was tested 20 times under realistic conditions. The results were recorded and analyzed through statistical

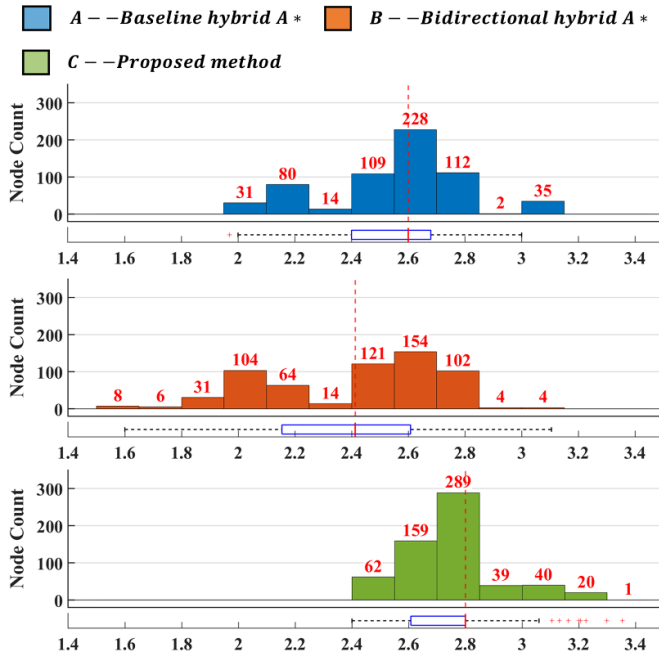


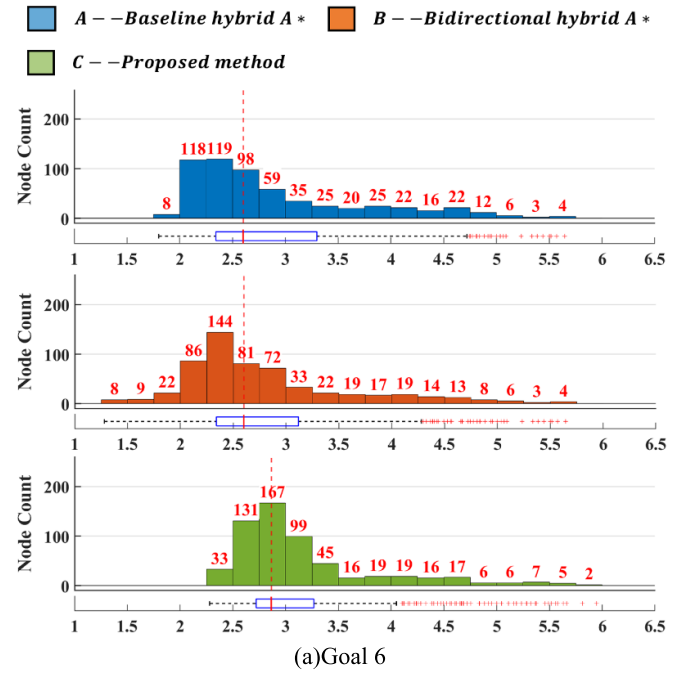
Figure 14. Safety margin distributions in single tunnel.

summaries of average search time and standard deviation (tables 7–9) and corresponding box plots (figures 17–19).

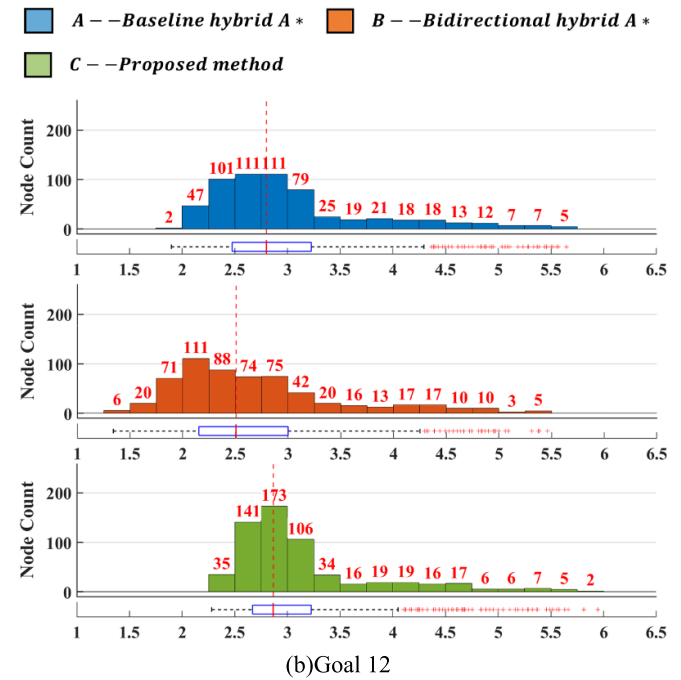
The experimental results indicate that the baseline Hybrid A* exhibits exponential growth in search time as the complexity of the search space increases, with peak durations reaching the order of several seconds. The bidirectional Hybrid A* improves performance, showing a slower rate of increase in moderately complex environments; however, it still incurs significant growth and displays numerous outliers in complex multi-branch and grid-like tunnels. In contrast, the proposed algorithm consistently maintains low search times (<35 ms), with a smooth, near-linear increase in time as environmental complexity rises. Notably, by reporting both the mean and standard deviation for each scenario, the results show that the proposed method not only achieves the shortest average search duration but also demonstrates the smallest standard deviations in almost all cases, indicating excellent consistency and robustness. Moreover, the method exhibits virtually no outliers across repeated trials, demonstrating superior real-time performance and robustness.

The number of expanded nodes provides an intuitive measure of the state exploration cost in path planning and is a critical indicator of the algorithm's search efficiency and spatial complexity [40]. It also serves as a direct factor influencing overall search time. By comparing the average number of expanded nodes across different goal combinations in three typical tunnel scenarios (as shown in figure 20), several observations can be made:

In all tested environments, the baseline Hybrid A* algorithm exhibits exponential growth in node expansion as the complexity of the environment increases. For instance, in the single tunnel scenario, the number of nodes increases from 14 to 9,463, and in the grid-like tunnel, it escalates dramatically to 287,522. The bidirectional Hybrid A* benefits



(a) Goal 6



(b) Goal 12

Figure 15. Safety margin distribution in the multi-branch tunnel.

from a dual search mechanism, which helps reduce redundant expansion in general, but it still suffers from a significant increase in node count for longer paths in multi-branch scenarios (e.g. goals 7–12) and the grid map. In contrast, the proposed algorithm consistently maintains a low number of expanded nodes (<300) across all scenarios, with negligible growth as environmental complexity rises. This reflects its strong efficiency and robust search behavior.

The visualizations of the **node expansion processes** along representative planned paths (as shown in figures 21 and 23) reveal distinct differences in search behavior across the three

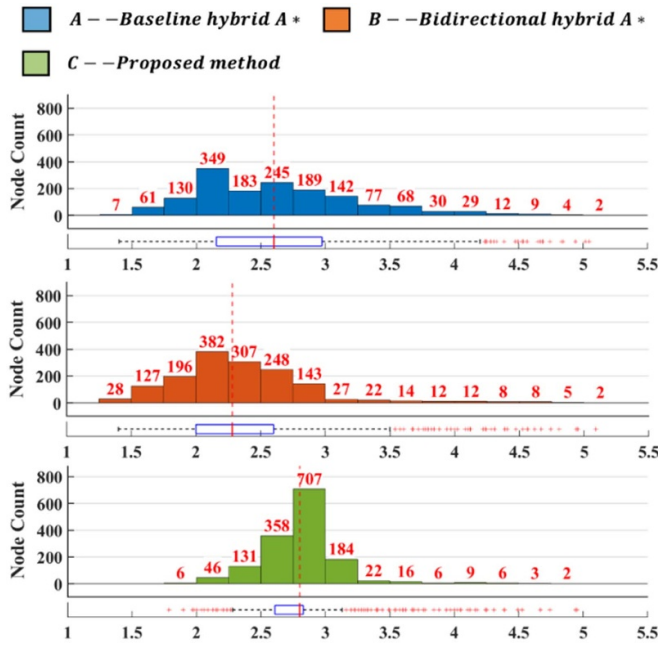


Figure 16. Safety margin distribution in the grid-like tunnel.

Table 7. Average searching duration in the single-tunnel scenario.

Index	Average searching duration (ms)		
	HA	BHA	HHA
1	0.61 ± 0.02	0.75 ± 0.02	9.13 ± 0.16
2	1.36 ± 0.05	1.50 ± 0.03	9.17 ± 0.14
3	4.62 ± 0.09	2.20 ± 0.03	9.29 ± 0.10
4	6.70 ± 0.13	2.90 ± 0.02	9.47 ± 0.12
5	11.96 ± 0.16	3.46 ± 0.07	9.48 ± 0.13
6	16.29 ± 0.33	4.06 ± 0.10	4.67 ± 0.10
7	17.36 ± 0.14	5.11 ± 0.10	9.63 ± 0.17
8	23.58 ± 0.21	5.45 ± 0.09	10.25 ± 0.27
9	30.69 ± 0.84	8.63 ± 0.12	5.27 ± 0.24
10	47.66 ± 0.52	7.52 ± 0.13	5.37 ± 0.32
11	59.15 ± 0.53	14.27 ± 0.41	10.48 ± 0.36
12	79.44 ± 0.52	10.12 ± 0.36	5.46 ± 0.21
13	99.05 ± 1.04	9.22 ± 0.10	10.50 ± 0.24
14	205.74 ± 1.87	9.87 ± 0.16	5.54 ± 0.17

Note: Bolded mean values indicate a shorter searching duration, while bolded standard deviations denote a more stable distribution of searching durations.

algorithms. The baseline Hybrid A* consistently exhibits widespread and sparsely distributed redundant expansions, with expansion directions diverging significantly, leading to inefficient search. The bidirectional Hybrid A* shows relatively more converged expansion patterns, yet still suffers from considerable redundancy, especially in complex road segments. In contrast, the proposed method demonstrates highly concentrated and well-aligned expansion patterns, with nodes densely distributed along the reference state set. This guided expansion effectively suppresses invalid search in complex environments and significantly improves planning efficiency and convergence.

Table 8. Average searching duration in the multi-branch tunnel scenario.

Index	Average searching duration (ms)		
	HA	BHA	HHA
1	9.97 ± 0.10	2.53 ± 0.26	5.12 ± 0.26
2	7.20 ± 0.09	8.07 ± 0.09	11.33 ± 0.30
3	9.22 ± 0.13	3.83 ± 0.12	12.64 ± 0.60
4	11.42 ± 0.15	15.29 ± 0.34	13.43 ± 0.23
5	9.20 ± 0.14	7.69 ± 0.20	14.41 ± 0.38
6	10.54 ± 0.10	7.31 ± 0.93	15.48 ± 0.27
7	17.30 ± 0.22	8.22 ± 0.86	10.46 ± 0.24
8	407.57 ± 2.25	11.53 ± 0.76	11.54 ± 0.23
9	862.77 ± 7.48	10.96 ± 0.09	7.69 ± 0.45
10	1722.19 ± 13.96	8.48 ± 0.34	13.63 ± 0.34
11	4582.63 ± 23.04	78.75 ± 0.89	14.54 ± 0.25
12	9324.35 ± 56.42	134.82 ± 1.81	15.89 ± 0.32

Note: Bolded mean values indicate a shorter searching duration, while bolded standard deviations denote a more stable distribution of searching durations.

Table 9. Average searching duration in the grid-like tunnel scenario.

Average searching duration (ms)		
HA	BHA	HHA
5110.25 ± 25.21	726.06 ± 9.62	34.86 ± 1.28

Note: Bolded mean values indicate a shorter searching duration, while bolded standard deviations denote a more stable distribution of searching durations.

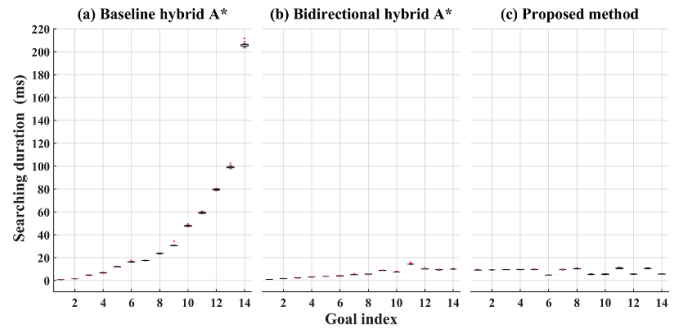


Figure 17. Searching duration distribution in the single-tunnel scenario.

A comprehensive comparison of search duration and node expansion strategies reveals clear differences in planning efficiency across the three methods. The proposed hierarchical path planning algorithm consistently exhibits superior performance and robustness across various constrained underground mining scenarios. Compared to the baseline and bidirectional Hybrid A* algorithms, the proposed method benefits from a structured reference state set and a region-specific dynamic expansion strategy, which together enable focused exploration and effectively suppress redundant node generation. This results in significantly shorter planning times and more consistent search behavior, demonstrating the method's efficiency and reliability under complex environmental constraints.

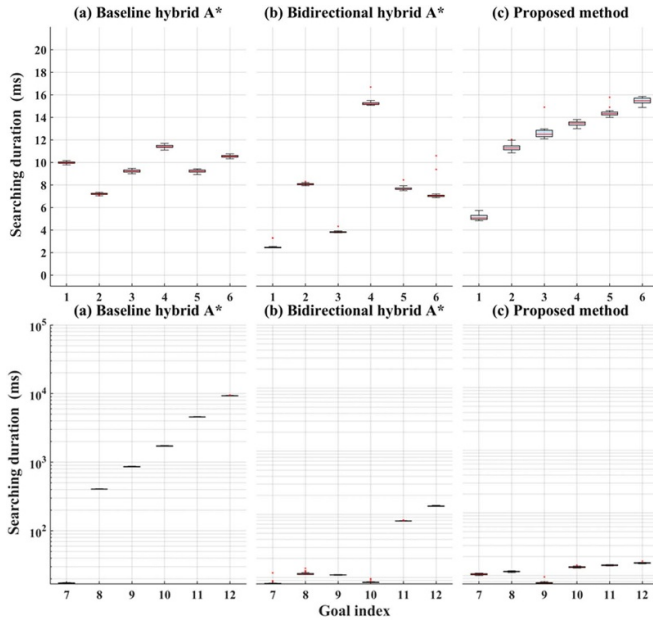


Figure 18. Search duration distribution in the multi-branch tunnel scenario.

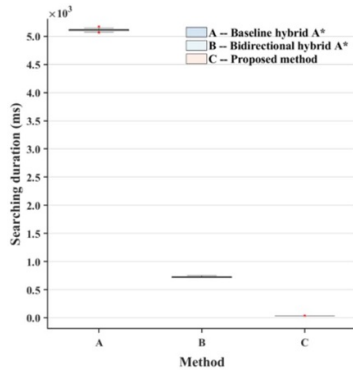


Figure 19. Search duration distribution in the grid-like tunnel scenario.

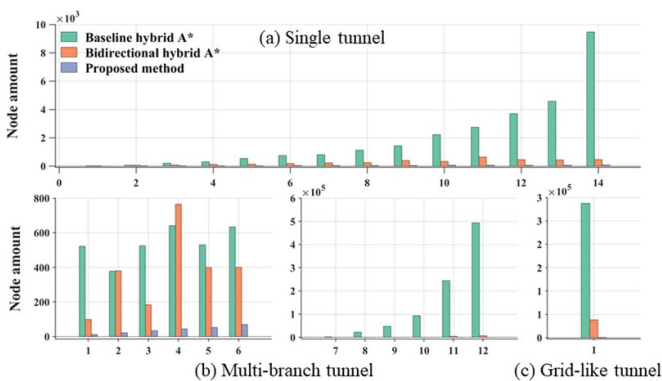


Figure 20. Node amount in all experimental tunnels.

Given the slow steering response and limited turning radius of underground LHD vehicles, path curvature plays a critical role in determining trajectory feasibility and tracking stability. As a supplementary metric to safety evaluation, the

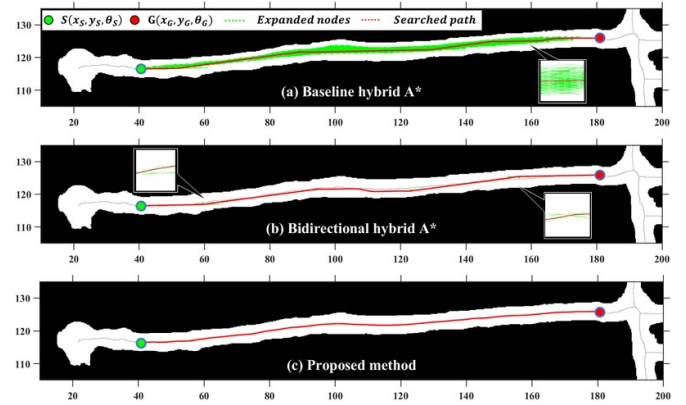


Figure 21. Illustration of node expansion in the single-tunnel scenario.

average curvature quantifies the geometric smoothness and steering load of planned trajectories. In this study, all planned paths were smoothed using cubic splines and uniformly resampled before computing the mean and standard deviation of curvature across all goal combinations in each scenario. Curvature distribution patterns along representative trajectories were further visualized via heatmaps.

As summarized in table 10–12, the baseline Hybrid A* consistently achieved the lowest average curvature across all scenarios, indicating superior path smoothness. The bidirectional Hybrid A* ranked second, where its bidirectional stitching introduced moderate curvature plateaus near connection points. Although the proposed method outperformed in terms of search efficiency and safety margin, it exhibited higher overall curvature due to the influence of reference-state tracking and current cost function weighting, often resulting in sharp curvature peaks near junctions and terminal alignments.

Heatmap visualizations (figures 24–26), further highlight the differences in smoothness among the three methods across different scenarios. The baseline Hybrid A* consistently produces the lowest curvature, with structured trajectories and concentrated turning points, showing only isolated peaks where necessary. The bidirectional Hybrid A* exhibits slightly higher curvature, introducing occasional sharp turns and continuous medium-curvature sections due to stitching, particularly in multi-branch and grid-like scenarios with repeated small-radius transitions. The proposed method, while achieving better alignment with the skeleton, tends to generate sustained medium-strength turns in the later segments, with pronounced curvature spikes during branch entry and final alignment. This results in lighter background tones and localized bright peaks, especially in grid-like scenarios where high-curvature segments are most evident near the start and end, indicating relatively lower geometric smoothness overall.

In summary, while the proposed method demonstrates superior performance in search efficiency and safety margin, it is slightly inferior in terms of path smoothness. In structurally complex regions, it tends to exhibit abrupt curvature changes and cumulative lateral deviations, resulting in relatively lower geometric continuity along the trajectory.

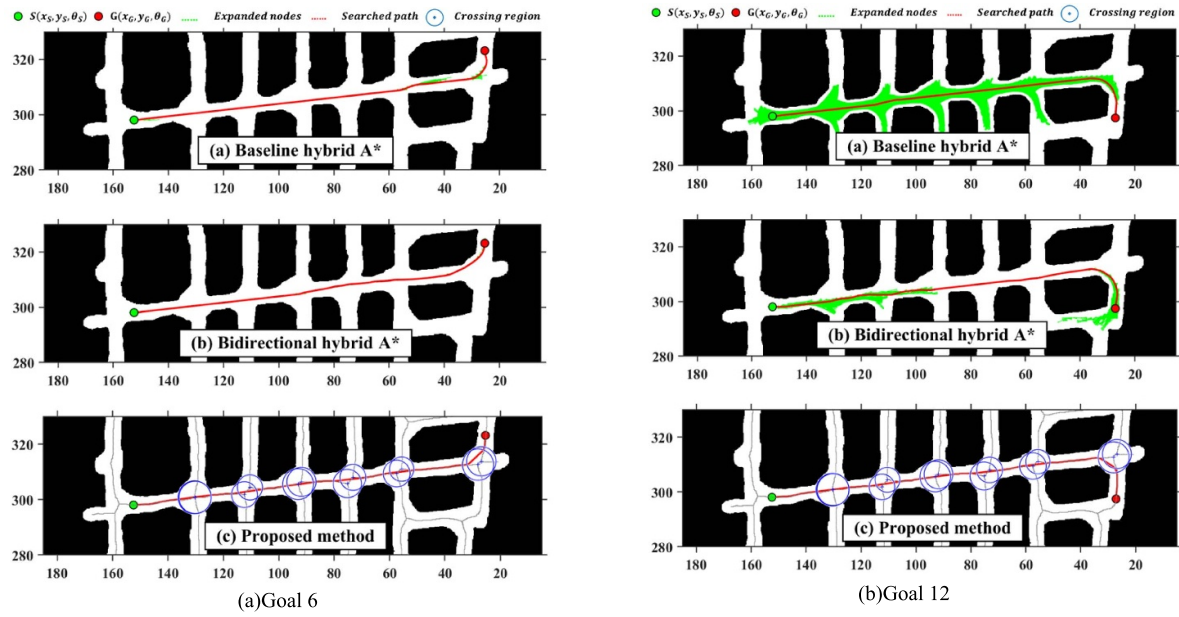


Figure 22. Illustration of node expansion in the multi-branch tunnel scenario.

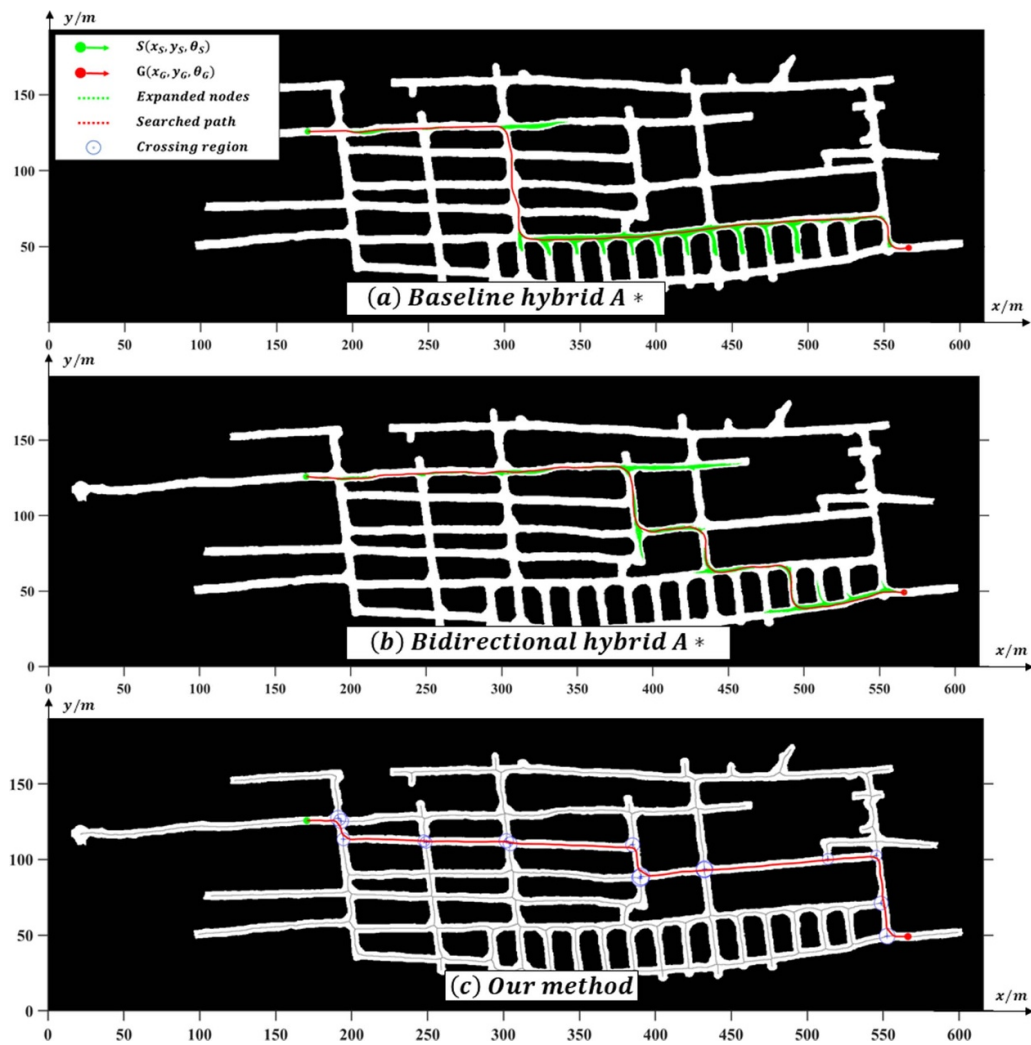


Figure 23. Illustration of node expansion in the grid-like tunnel scenario.

Table 10. Average curvature of paths in the single-tunnel scenario.

Index	Average curvature		
	HA	BHA	HHA
1	$4.4 \times 10^{-4} \pm 4.2 \times 10^{-4}$	$3.7 \times 10^{-4} \pm 4.0 \times 10^{-4}$	$3.4 \times 10^{-2} \pm 5.4 \times 10^{-2}$
2	$1.0 \times 10^{-2} \pm 2.9 \times 10^{-2}$	$5.2 \times 10^{-3} \pm 9.2 \times 10^{-3}$	$5.7 \times 10^{-2} \pm 5.4 \times 10^{-2}$
3	$1.3 \times 10^{-2} \pm 2.6 \times 10^{-2}$	$8.5 \times 10^{-3} \pm 2.3 \times 10^{-2}$	$4.5 \times 10^{-2} \pm 4.4 \times 10^{-2}$
4	$1.0 \times 10^{-2} \pm 2.5 \times 10^{-2}$	$2.0 \times 10^{-2} \pm 2.9 \times 10^{-2}$	$3.6 \times 10^{-2} \pm 3.6 \times 10^{-2}$
5	$8.2 \times 10^{-3} \pm 2.2 \times 10^{-2}$	$7.9 \times 10^{-3} \pm 1.8 \times 10^{-2}$	$3.2 \times 10^{-2} \pm 2.8 \times 10^{-2}$
6	$8.2 \times 10^{-3} \pm 2.5 \times 10^{-2}$	$1.1 \times 10^{-2} \pm 2.3 \times 10^{-2}$	$2.7 \times 10^{-2} \pm 2.4 \times 10^{-2}$
7	$7.5 \times 10^{-3} \pm 2.0 \times 10^{-2}$	$1.3 \times 10^{-2} \pm 2.6 \times 10^{-2}$	$2.7 \times 10^{-2} \pm 2.4 \times 10^{-2}$
8	$6.2 \times 10^{-3} \pm 1.8 \times 10^{-2}$	$6.1 \times 10^{-3} \pm 1.8 \times 10^{-2}$	$2.7 \times 10^{-2} \pm 2.3 \times 10^{-2}$
9	$5.9 \times 10^{-3} \pm 1.8 \times 10^{-2}$	$1.9 \times 10^{-2} \pm 3.1 \times 10^{-2}$	$2.3 \times 10^{-2} \pm 2.0 \times 10^{-2}$
10	$8.6 \times 10^{-3} \pm 2.2 \times 10^{-2}$	$1.3 \times 10^{-2} \pm 2.5 \times 10^{-2}$	$2.0 \times 10^{-2} \pm 1.7 \times 10^{-2}$
11	$9.0 \times 10^{-3} \pm 2.4 \times 10^{-2}$	$1.0 \times 10^{-2} \pm 2.3 \times 10^{-2}$	$2.1 \times 10^{-2} \pm 1.8 \times 10^{-2}$
12	$8.1 \times 10^{-3} \pm 2.1 \times 10^{-2}$	$9.4 \times 10^{-3} \pm 2.2 \times 10^{-2}$	$1.9 \times 10^{-2} \pm 1.5 \times 10^{-2}$
13	$7.9 \times 10^{-3} \pm 2.1 \times 10^{-2}$	$6.0 \times 10^{-3} \pm 1.8 \times 10^{-2}$	$1.9 \times 10^{-2} \pm 1.5 \times 10^{-2}$
14	$1.0 \times 10^{-2} \pm 2.3 \times 10^{-2}$	$1.0 \times 10^{-2} \pm 2.5 \times 10^{-2}$	$1.7 \times 10^{-2} \pm 1.3 \times 10^{-2}$

Note: Bolded mean values indicate a smaller average curvature, while bolded standard deviations denote a smoother distribution of curvature.

Table 11. Average curvature of paths in the multi-branch tunnel scenario.

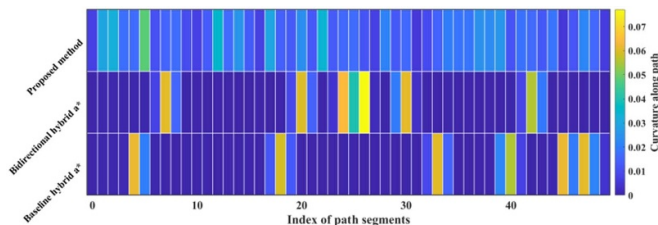
Index	Average curvature		
	HA	BHA	HHA
1	$5.1 \times 10^{-2} \pm 5.9 \times 10^{-2}$	$6.5 \times 10^{-2} \pm 4.7 \times 10^{-2}$	$6.7 \times 10^{-2} \pm 8.7 \times 10^{-2}$
2	$3.2 \times 10^{-2} \pm 5.3 \times 10^{-2}$	$4.7 \times 10^{-2} \pm 4.4 \times 10^{-2}$	$5.6 \times 10^{-2} \pm 7.7 \times 10^{-2}$
3	$2.5 \times 10^{-2} \pm 4.6 \times 10^{-2}$	$3.6 \times 10^{-2} \pm 4.5 \times 10^{-2}$	$5.2 \times 10^{-2} \pm 7.2 \times 10^{-2}$
4	$2.0 \times 10^{-2} \pm 4.5 \times 10^{-2}$	$3.5 \times 10^{-2} \pm 4.3 \times 10^{-2}$	$4.8 \times 10^{-2} \pm 7.1 \times 10^{-2}$
5	$1.6 \times 10^{-2} \pm 4.1 \times 10^{-2}$	$3.0 \times 10^{-2} \pm 4.0 \times 10^{-2}$	$5.2 \times 10^{-2} \pm 8.0 \times 10^{-2}$
6	$2.2 \times 10^{-2} \pm 5.0 \times 10^{-2}$	$2.3 \times 10^{-2} \pm 3.6 \times 10^{-2}$	$4.5 \times 10^{-2} \pm 6.5 \times 10^{-2}$
7	$6.2 \times 10^{-2} \pm 7.1 \times 10^{-2}$	$7.1 \times 10^{-2} \pm 9.2 \times 10^{-2}$	$8.2 \times 10^{-2} \pm 8.9 \times 10^{-2}$
8	$5.3 \times 10^{-2} \pm 6.1 \times 10^{-2}$	$5.5 \times 10^{-2} \pm 5.3 \times 10^{-2}$	$6.0 \times 10^{-2} \pm 7.8 \times 10^{-2}$
9	$2.8 \times 10^{-2} \pm 5.3 \times 10^{-2}$	$4.7 \times 10^{-2} \pm 4.4 \times 10^{-2}$	$6.0 \times 10^{-2} \pm 7.9 \times 10^{-2}$
10	$2.7 \times 10^{-2} \pm 4.5 \times 10^{-2}$	$4.0 \times 10^{-2} \pm 4.3 \times 10^{-2}$	$4.9 \times 10^{-2} \pm 7.3 \times 10^{-2}$
11	$2.3 \times 10^{-2} \pm 4.3 \times 10^{-2}$	$3.4 \times 10^{-2} \pm 4.4 \times 10^{-2}$	$4.7 \times 10^{-2} \pm 7.0 \times 10^{-2}$
12	$1.8 \times 10^{-2} \pm 4.0 \times 10^{-2}$	$3.7 \times 10^{-2} \pm 4.4 \times 10^{-2}$	$4.6 \times 10^{-2} \pm 6.6 \times 10^{-2}$

Note: Bolded mean values indicate a smaller average curvature, while bolded standard deviations denote a smoother distribution of curvature.

Table 12. Average curvature of paths in the grid-like tunnel scenario.

Index	Average curvature		
	HA	BHA	HHA
1	$2.5 \times 10^{-2} \pm 4.1 \times 10^{-2}$	$4.5 \times 10^{-2} \pm 4.5 \times 10^{-2}$	$5.3 \times 10^{-2} \pm 6.7 \times 10^{-2}$

Note: Bolded mean values indicate a smaller average curvature, while bolded standard deviations denote a smoother distribution of curvature.

**Figure 24.** Curvature along paths in the single-tunnel scenario.

5. Discussion and future work

This study conducted simulation-based evaluations comparing the proposed method with the baseline Hybrid A* and its

bidirectional variant, focusing on key performance metrics including path length, safety margin, node expansion efficiency, and path smoothness. The results demonstrate that the proposed approach offers strong adaptability and search performance in complex underground tunnel environments. Nonetheless, several limitations remain, and further research is required in the following aspects to enhance the algorithm's robustness and applicability.

To begin with, the current multi-objective heuristic cost function includes several critical parameters—such as penalties for path length, steering dynamics, and articulation angle stability—which are determined through empirical tuning and extensive trial-and-error. While the selected weights yield satisfactory results in controlled scenarios, the lack of

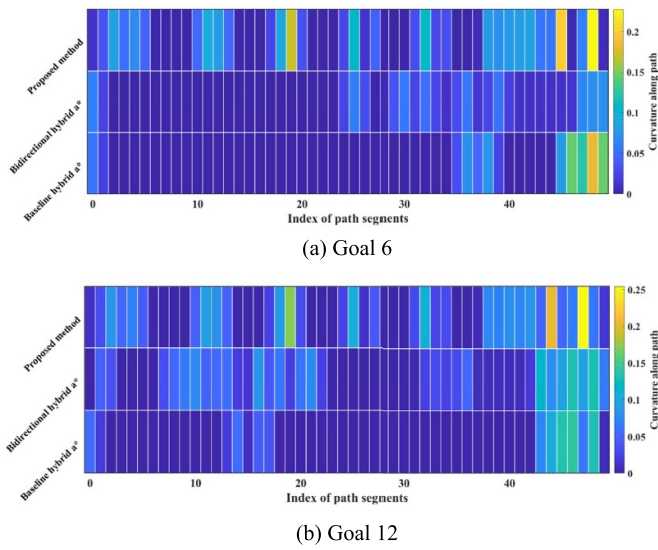


Figure 25. Curvature along paths in the multi-branch tunnel scenario.

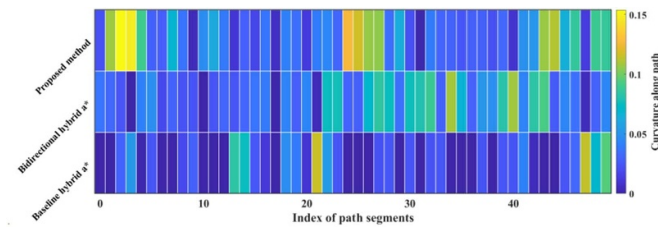


Figure 26. Curvature along paths in the grid-like tunnel scenario.

systematic quantitative analysis and the high cost of manual tuning limit scalability. Moreover, the generalizability of these fixed parameters across different tunnel structures or LHD vehicle models remains uncertain. Future work will explore reinforcement learning and optimization-based strategies to enable adaptive cost weight adjustment, thereby improving algorithm generalization and engineering applicability.

Another critical challenge lies in the generation of the reference state set relies on the environment map constructed via SLAM and the corresponding GVD extracted from it. However, in real-world applications, underground tunnel environments are highly unstructured, and irregular wall boundaries can compromise the quality of the GVD skeleton. To improve the robustness of reference generation, future work may incorporate map pre-processing and passable region segmentation strategies, combined with confidence-aware pruning or redundant path extraction methods, to enhance the structural completeness and spatial stability of the GVD.

Moreover, the current algorithm operates on a static global map and does not account for dynamic obstacles that frequently appear in mining operations, such as mobile machinery, workers, and temporary debris. These dynamic elements impose higher demands on real-time responsiveness and planning robustness. To address this, future extensions will incorporate incremental map updates and integrate local

obstacle avoidance strategies—such as model predictive control or the dynamic window approach—to form a global-local collaborative planning framework capable of real-time perception and adaptive path replanning.

In addition, the current planner assumes a two-dimensional operational environment, without modeling elevation variations or multi-level spatial structures commonly found in underground mines. This simplification may affect vehicle stability, energy efficiency, and introduce unforeseen challenges to search strategies. Future work will incorporate elevation data and slope constraints into the planning process, enhancing path feasibility judgment and cost modeling to improve adaptability and reliability in realistic 3D mining environments.

Despite these advantages, abrupt curvature changes are still observed at the transitions between reference segments and *Dubins* curves—particularly during segment stitching and terminal alignment. These geometric discontinuities may increase control complexity and reduce motion smoothness. Future research will explore curvature-continuous connection strategies, such as clothoid-based transitions or higher-order spline interpolation, to improve path geometric smoothness and execution feasibility.

Finally, the proposed method has not yet been deployed or validated on real LHD platforms. In practical applications, factors such as perception noise, control delay, and communication instability may degrade system performance and operational reliability. As a next step, we plan to modularize the algorithm for seamless integration with existing autonomous mining platforms, conduct hardware-in-the-loop validation, and ultimately carry out field trials in real underground mining environments to comprehensively assess its engineering feasibility and runtime stability.

6. Conclusion

This paper presents a hierarchical path planning algorithm tailored for constrained underground mining environments. By integrating the A* algorithm with a GVD, the proposed method constructs an upper-level reference state set aligned with tunnel centerlines. Combined with an improved node expansion strategy and a multi-objective heuristic cost function, the algorithm effectively generates feasible paths that respect the nonholonomic motion constraints of articulated vehicles such as LHD machines. Compared with the baseline Hybrid A* and bidirectional Hybrid A* algorithms, the proposed method demonstrates significant advantages in key performance indicators, including path length control, safety margin assurance, and planning efficiency. In particular, it exhibits its superior robustness and computational stability in complex environments.

Experimental comparisons across three representative underground tunnel scenarios verify that the proposed method achieves an optimal balance between path feasibility and search efficiency by actively guiding the search direction and effectively suppressing redundant expansions. These findings highlight its strong potential for improving autonomous path planning capabilities of LHD vehicles in confined spaces, and

provide a generalizable framework for multi-constraint path generation in structured environments. Future work will focus on further optimizing the construction of the reference state set and enhancing path smoothness and geometric quality to improve the overall trajectory performance.

Data availability statement

The data cannot be made publicly available upon publication because they contain commercially sensitive information. The data that support the findings of this study are available upon reasonable request from the authors.

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